Customer Churn Analysis and Prediction

Task 2

Exploratory Data Analysis (EDA)

Internship Report

Dataset: Telco Customer Churn Dataset

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1. Introduction

This report covers Task 2 of the Customer Churn Analysis and Prediction project: Exploratory Data Analysis (EDA). Working from the cleaned and encoded dataset produced in Task 1, the objective here was to measure the overall churn rate, understand how churn relates to customer demographics, and investigate how churn varies across tenure, contract type, payment method, and other service-related attributes. Each section below presents the business question I asked, the chart or table I produced to answer it, and the interpretation of the result.

2. Overall Churn Rate

Business question: What percentage of customers left the company?

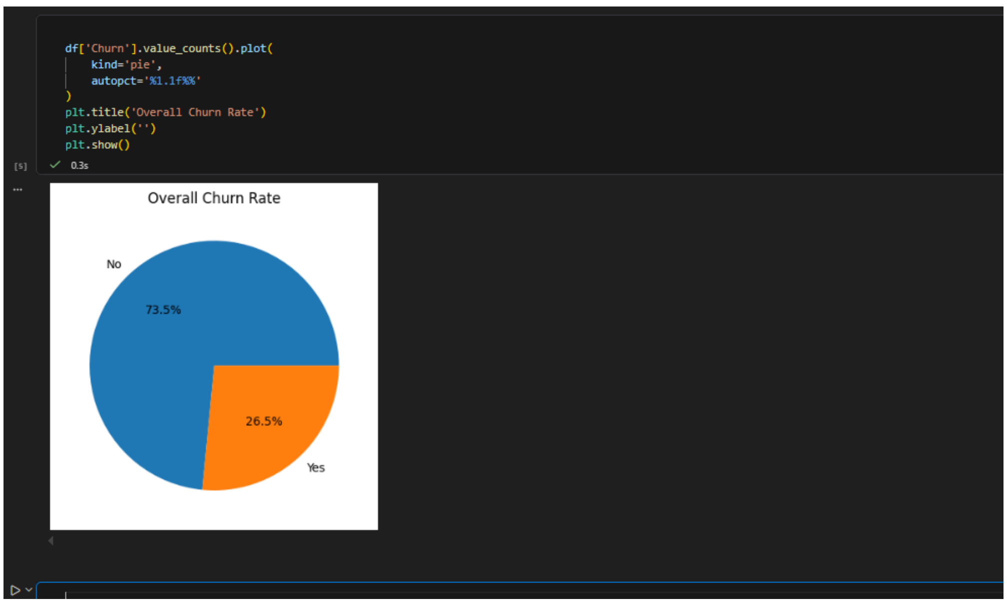

Figure 1. Overall churn rate — share of retained versus churned customers.

The overall churn rate was calculated to measure the proportion of customers who discontinued the service. The results show that 26.54% of customers churned, while 73.46% remained with the company. This means that roughly one out of every four customers left the service, which represents a considerable loss and justifies a deeper investigation into what drives churn.

> **Answer**
>
> *Yes, customer churn is a real problem for the company: around 26.5% of customers leave, a substantial share that motivates the rest of this analysis.*

3. Customer Demographics and Churn

3.1 Gender

```python
gender_churn = pd.crosstab(
    df['gender'],
    df['Churn'],
    normalize='index'
) * 100

print(gender_churn)
```

```
Churn           No        Yes
gender
Female   73.079128  26.920872
Male     73.839662  26.160338
```

Figure 2. Churn rate by gender.

Female customers show a churn rate of 26.92%, compared to 26.16% for male customers. Since the difference is less than one percentage point, gender has very little influence on churn and is unlikely to be a useful predictor on its own.

3.2 Partner Status

```python
pd.crosstab(
    df['Partner'],
    df['Churn'],
    normalize='index'
) * 100
```

Churn	No	Yes
Partner		
No	67.042021	32.957979
Yes	80.335097	19.664903

Figure 3. Churn rate by partner status (crosstab, % by row).

Customers with a partner show a significantly lower churn rate (19.7%) compared to customers without a partner (33.0%). This suggests that customers with greater household stability are more likely to remain with the service. This feature is moderately useful and works best when combined with other features such as Contract and tenure.

3.3 Dependents

```python
pd.crosstab(df['Dependents'], df['Churn'], normalize='index') * 100
```

Churn	No	Yes
Dependents		
No	68.720860	31.279140
Yes	84.549763	15.450237

Figure 4. Churn rate by dependents status.

Customers with dependents show a significantly lower churn rate (15.5%) compared to those without dependents (31.3%). This indicates that customers with family responsibilities are more likely to stay loyal to the service, making Dependents a comparatively strong predictive feature — almost as informative as Contract and tenure, and useful for segmenting family versus non-family customers.

4. Contract Type Analysis

```
[15]
pd.crosstab(df['Contract'], df['Churn'], normalize='index') * 100
0.0s
```

Churn	No	Yes
Contract		
Month-to-month	57.290323	42.709677
One year	88.730482	11.269518
Two year	97.168142	2.831858

Figure 5. Churn rate by contract type.

There is a clear pattern: the longer the contract, the lower the churn. Contract type turned out to be the most significant factor influencing churn in the exploratory analysis. Customers on month-to-month contracts show a very high churn rate (42.7%), while customers on one-year and two-year contracts show much lower churn rates (11.3% and 2.8% respectively).

This happens because month-to-month customers have no commitment and can leave at any time, while one- and two-year contracts usually come with discounts and create a "lock-in" effect that makes cancelling less attractive. From a business perspective, this suggests focusing retention strategies on converting month-to-month customers to longer commitments, for example through yearly-plan discounts and stronger early retention efforts.

5. Payment Method Analysis

```
[16]
pd.crosstab(df['PaymentMethod'], df['Churn'], normalize='index') * 100
0.0s
```

Churn	No	Yes
PaymentMethod		
Bank transfer (automatic)	83.290155	16.709845
Credit card (automatic)	84.756899	15.243101
Electronic check	54.714588	45.285412
Mailed check	80.893300	19.106700

Figure 6. Churn rate by payment method.

Customers using electronic checks show a significantly higher churn rate (45.3%) compared to those using automatic payment methods such as credit cards (15.2%) and bank transfers (16.7%). A likely explanation is that electronic checks involve more manual friction and are often used by less tech-engaged or less loyal customers, whereas automated payments are associated with better retention. This suggests the company should encourage automatic payments, for instance through incentives for switching away from electronic checks.

6. Internet Service Type Analysis

```
pd.crosstab(df['InternetService'], df['Churn'], normalize='index') * 100
```

✓ 0.0s

Churn / InternetService	No	Yes
DSL	81.040892	18.959108
Fiber optic	58.107235	41.892765
No	92.595020	7.404980

Figure 7. Churn rate by internet service type.

Customers using fiber optic internet show a notably higher churn rate (41.9%) compared to DSL users (19.0%) and customers without internet service (7.4%). This is an important finding: even though fiber is a faster, more modern and premium service, it is also the highest-risk segment after electronic-check users. Possible explanations include higher pricing compared to DSL, higher customer expectations leading to more complaints, service-quality issues, or stronger competition on fiber offers. Fiber customers can therefore be described as high-value but high-risk.

7. Technical Support Analysis

```
pd.crosstab(df['TechSupport'], df['Churn'], normalize='index') * 100
```

✓ 0.0s

Churn / TechSupport	No	Yes
No	58.364526	41.635474
No internet service	92.595020	7.404980
Yes	84.833659	15.166341

Figure 8. Churn rate by technical support status.

Customers without technical support show a very high churn rate (41.64%), compared to 15.17% for customers with tech support, and only 7.40% for customers with no internet service at all (a different, low-dependency segment). This indicates that technical assistance plays an important role in customer satisfaction: faster problem resolution builds trust and reduces the frustration that leads customers to switch providers. The company could reduce churn by promoting tech-support plans and targeting no-support customers with retention campaigns.

8. Tenure Analysis

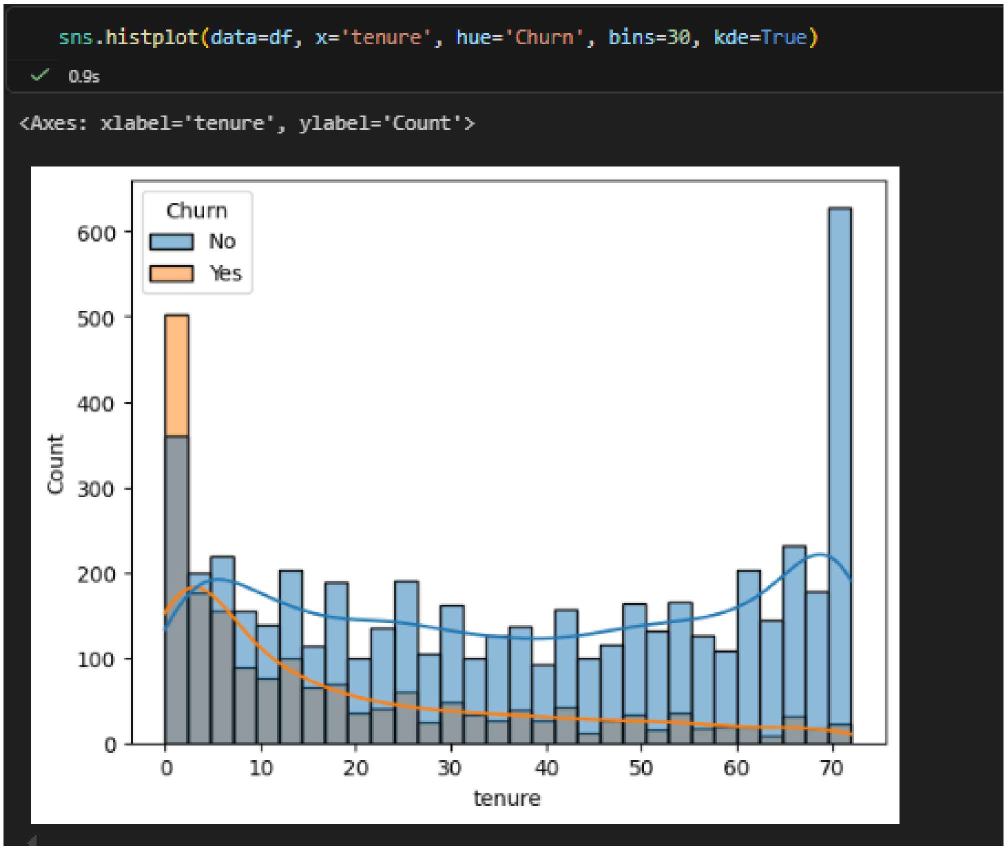

Figure 9. Churn distribution across customer tenure (months).

Plotting churn against tenure answers a key question: do long-term customers leave less? The analysis shows that churn is heavily concentrated among new customers: there is a large peak of churn between 0 and 10 months of tenure, after which the churn rate steadily decreases. Beyond roughly 20–30 months, churn becomes much lower, and at high tenure (50–70 months) almost no customers churn.

Tenure is therefore one of the strongest indicators of churn: the longer a customer stays, the less likely they are to leave, most likely because trust builds over time, switching becomes less convenient, and early dissatisfaction is the main driver of early departures. This points directly to the importance of onboarding: the company should focus on the first three months of the relationship, with early support and welcome offers to reduce early churn.

9. Monthly Charges Analysis

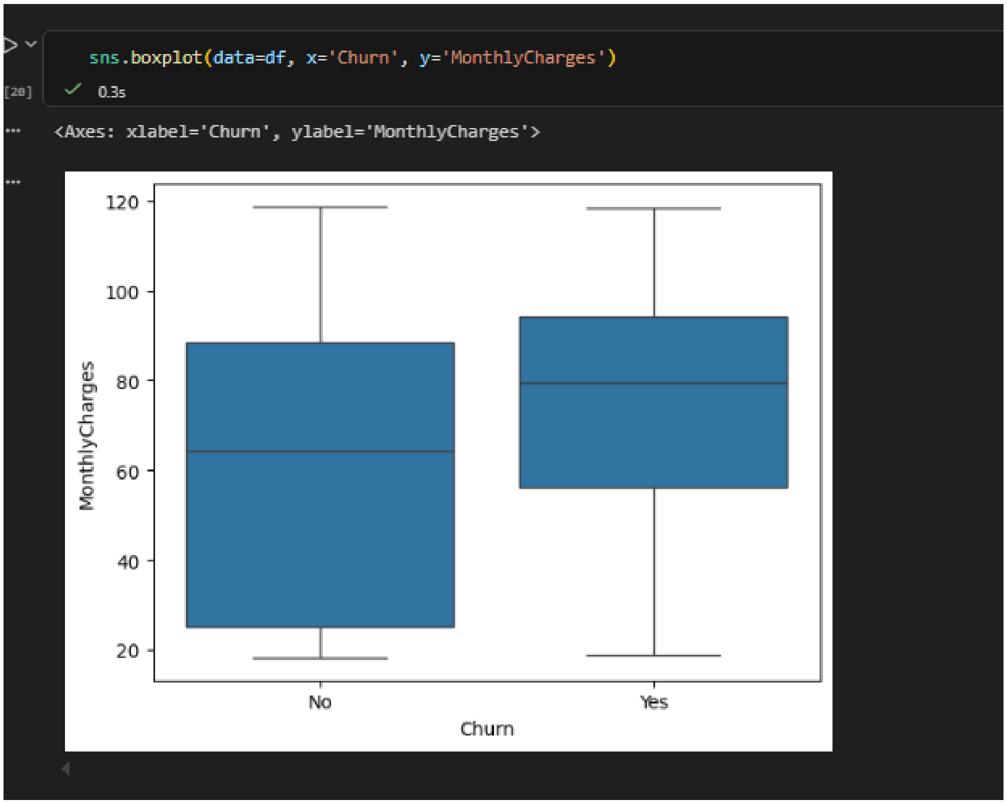

Figure 10. Distribution of monthly charges by churn status.

Comparing monthly charges between churned and retained customers answers another question: do high-paying customers churn more? The median monthly charge for churned customers (roughly 75–80) is clearly higher than for retained customers (roughly 60–65), and churned customers are concentrated in the higher-charge range, although some overlap remains — some low-paying customers still churn and some high-paying customers stay.

This points to a positive relationship between price and churn risk: higher monthly cost appears to increase the likelihood of leaving, possibly because customers feel overcharged, competitors offer cheaper plans, or high-paying customers simply have higher expectations. MonthlyCharges is therefore an important factor, but one that works best in combination with Contract and tenure rather than alone.

10. Key Insights Summary

- Overall churn rate is 26.5%, a significant share of the customer base.
- Gender has almost no effect on churn.
- Partner status and dependents both reduce churn, with dependents being the stronger of the two.
- Contract type is the single most powerful churn driver: month-to-month contracts churn far more than one- or two-year contracts.
- Electronic check users churn much more than customers on automatic payment methods.
- Fiber optic customers are high-value but high-risk, showing high churn despite being a premium service.
- Lack of technical support is strongly associated with higher churn.

- Churn is concentrated in the first few months of tenure and drops sharply for long-term customers.
- Higher monthly charges are associated with a higher probability of churn.

11. Conclusion

The exploratory analysis shows that churn at this company is not random: it is strongly concentrated among new, month-to-month, fiber-optic customers who pay by electronic check and lack technical support, especially within their first few months. These patterns give a clear, data-driven direction for the customer segmentation work in Task 3 and the predictive modeling carried out in Task 4.